2.2 Solving Inequalities Using Addition or Subtraction

Name _____ Period ____ Date _____

Algebra I 2.2 (30-minute plan) · Student sheet

I can…

- solve inequalities using addition.
- solve inequalities using subtraction and graph the solutions.
- write and solve inequalities for real-life problems.

Vocabulary · fill in the missing word

Addition Property of Inequality — _____ the same number to both sides keeps the inequality true

Subtraction Property of Inequality — subtracting the same number from both sides _____ the inequality true

Equivalent inequalities — inequalities with the same _____

Part 1 · Notes and examples (fill in each note, then work the example)

Addition and Subtraction Properties: Adding or subtracting the same number on both sides _____ change the symbol.

Check: Check one value _____ the solution and one _____ it.

Example 1

Solve $x - 6 \geq -10$.

Reading it backward: $4 \geq x$ means the same as x ____ 4.

Example 2

Solve $9 < y + 13$ and graph the solution.

Part 2 · Partner practice

Partner practice

Solve $m + 4.5 < 2$.